

UNIVERSITY OF ZAMBIA
SCHOOL OF HUMANITIES AND SOCIAL SCIENCES
DEPARTMENT OF ECONOMICS
ECN 2311 TUTORIAL SHEET 1 – 2023

QUESTION ONE

If U = set of all students at the University of Zambia (UNZA), A = set of all students who are in the first year at UNZA, B = set of all male students at UNZA, C = set of Economics students at UNZA,

- a. Show the following using a Venn diagram:
 - i. $A - B$;
 - ii. $B' \cap C$
 - iii. $A \cap B'$
 - iv. $A \cap (B \cap C)$
- b. If $A = \{2, 3, 4, 5\}$ $B = \{4, 5, 6, 7\}$ $C = \{6, 7, 8, 9\}$ $D = \{8, 9, 10, 11\}$, find
 - i. $A \cap B$
 - ii. $A \cup C$
 - iii. $(A \cup B) \cup C$
 - iv. $A \cap (B \cup C)$

QUESTION TWO

- a) Define a set.
- b) Given sets A and B , how can you prove the equivalence of these two sets?
- c) Suppose there are three newspapers published in Lusaka: The Sun, the Moon, and the Star. It has been found that 35% of the town's population read the Sun, 42% read the Moon, 51 % read the Star, 4% read the Sun and the Moon only, 10% read the Sun and the Star only, 15% read the Moon and the Star only, and 7% of the population reads all three newspapers.
 - i. Draw a Venn diagram to represent this information. Label A , the set that represents the Sun readers, B the set that represents the Moon readers, and C the set that represents the Star readers.
 - ii. What percentage of the population does not read any of the three newspapers?
 - iii. Find the percentage of the population that reads exactly one newspaper.
 - iv. Find the percentage of the population that reads the Sun or the Moon but not the Star.

- v. A local radio station states that 71 % of the population reads either the Moon or the Star. Use your Venn diagram to decide whether the statement is true and justify your answer.

QUESTION THREE

A survey was carried out by the UNZABECA Finance Committee to find out which electives they prefer. Three courses they chose from were: Money and Banking, Public Finance, and Corporate Finance. By letting A represents Money and Banking, B Public Finance, and C Corporate Finance respectively.

The following results were obtained: The total number of students interviewed was 14,520. Of these, course A was preferred by 8150 students, course B was preferred by 5540 and course C by 4010. Courses A and B were preferred by 3150 students. Courses B and C were preferred by 1820 students and those who preferred A and C were 1030. The students who did not prefer any of the three courses were 2520.

Find:

- a. The number of students who liked all the three courses.
- b. The number of courses who liked A but not B.
- c. The number of courses who liked A only.

QUESTION FOUR

a. Evaluate the following problems:

i. $\lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$

ii. $\lim_{x \rightarrow \infty} \left(\frac{6x+1}{2x+5} \right)$

iii. $\lim_{x \rightarrow 3} \left(\frac{x^2-9}{x-3} \right)$

b. Find the limit of the following:

i. $\lim_{t \rightarrow 2} \left(\frac{t^2-4}{t-2} \right)$

ii. $\lim_{x \rightarrow 5} \left(\frac{x-5}{x^2-25} \right)$

QUESTION FIVE

Given the following functions:

a. $f(x) = x^2 - 3x$

b. $f(x) = \begin{cases} x^2, & x \leq 2. \\ 5 - x, & x > 2. \end{cases}$

c. $f(x) = \frac{x+4}{x-2}, x \neq 2$

- i. Sketch the functions for the domain $(-1, 5)$
- ii. Determine the limit and continuity of the functions at $x=2$.

QUESTION SIX

Differentiate the following functions:

a. $y = 2 + 5x$

b. $y = \frac{x^4 + 2}{x^2 + 5x}$

c. $y = (2x^3 + 4x)^{\frac{1}{2}}$

d. $y = x \ln x - x$

e. $y = e^{(2x^2 - 3)}$

- f. For the function below, find the derivate with respect to x

$$f(x) = \left(\frac{3x - 1}{2x + 5} \right)^3$$

QUESTION SEVEN

- a. Find $\partial y / \partial x$ and $\partial y / \partial z$ when

i. $y = 6 + 3x + 16z + 4x^2 + 2z^2$

ii. $y = 14x^3z^2$

iii. $y = 9 + 4xz - 3x - 2z^3$

- b. Derive the four second-order partial derivatives for the production function:

$$Q = 6K + 0.3K^2L + 1.2L^2$$